DIPESH GHIMIRE

Computer Science & IT Student | Python & Django Developer | Aspiring AI/ML Engineer

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SUMMARY

8th-semester B.Sc. CSIT student with practical experience developing a team-based web platform using Python and Django, and an end-to-end machine learning project for early sepsis prediction using XGBoost and Streamlit. Comfortable with Python programming, database-driven application development, and integrating external services such as OpenStreetMap. Currently developing foundational skills in Data Science and Machine Learning, including regression, classification, statistics, NumPy, Pandas, and Matplotlib, reinforced by hands-on Kaggle courses and multiple job simulations. Interested in applying software engineering skills to AI/ML, NLP, computer vision, and intelligent document-processing systems.

EDUCATION

B.Sc. Computer Science and Information Technology (B.Sc. CSIT) | *Nihareeka College, affiliated with Tribhuvan University*
Expected Graduation: 2083

PROJECTS

Sepsis Prediction Using Machine Learning — Internship Project | *Python, XGBoost, Scikit-learn, SHAP, Streamlit*

- Built an **end-to-end ML pipeline** for early sepsis prediction on the **PhysioNet/CinC 2019** ICU dataset (**1.55M rows, 40,336 patients**).
- Prevented leakage with **patient-level splits** and patient-wise forward filling; handled **~1.8% class imbalance**.
- Engineered **5 clinical domain scores**; trained **XGBoost** and compared against **4 baseline models**.
- Achieved the best test **ROC-AUC 0.791** and **PR-AUC 0.093**; explained results with **SHAP**.
- Built a **Streamlit app** that returns a sepsis probability from clinical inputs.

GitHub: github.com/DipeshGhimire33/Intern_project

Nepal Cybercrime Data Analysis | *Python, Pandas, Plotly, Matplotlib, Streamlit*

- Analyzed **reported cybercrime in Nepal** across **3 fiscal years** (**9,013 → 19,730 → 18,926**) by medium, victim group & province.
- Cleaned data and engineered features (**year-over-year change**, **medium categories**, province–demographic joins) across **5 notebooks**.
- Ran **cross-dataset validation**: totals matched in all years except a **5-record gap in 2022/23**.
- Identified **social media (mainly Facebook)** as the top medium; found data **unsuitable for ML** (aggregate, 3 years).
- Built an interactive **Streamlit + Plotly dashboard** with 7 sections.

GitHub: github.com/DipeshGhimire33/Machine_learning_Projects_Compilation/tree/main/Cyber_crime

Tourist Platform — Team Project | *Python, Django, SQLite, HTML, CSS, JavaScript, OpenStreetMap*

- Developed a **travel-planning web platform** with a **team of 3**, bringing together vehicles, equipment, guides, and hotels.
- Built the backend with **Python/Django** and **SQLite**, including **Django authentication**.
- Integrated **OpenStreetMap** for route and location features.
- Handled **database migrations** and schema changes while collaborating on a shared codebase.

GitHub: github.com/DipeshGhimire33/Tourist_travels_7th_sem_project

TECHNICAL SKILLS

Programming: Python, C, C++

Web Development: Django, HTML, CSS, JavaScript

Data & Scientific Computing: NumPy, Pandas, Matplotlib, Seaborn

Machine Learning: Regression, Classification, XGBoost, Scikit-learn, SHAP, Machine Learning Fundamentals

Application Development: Streamlit

Database: SQLite, SQL fundamentals

Other: Git/GitHub, Jupyter Notebook, REST/API concepts, OpenStreetMap integration

DATA SCIENCE & MACHINE LEARNING

Currently completing a 100-day Data Science & Machine Learning program

- Building foundational knowledge in machine learning and data analysis.
- Familiar with regression and classification concepts, applied in a hands-on imbalanced-classification project.
- Developing practical experience with NumPy, Pandas, and Matplotlib.
- Academic background in statistics through B.Sc. CSIT coursework.
- Currently expanding knowledge toward NLP, AI/ML applications, and real-world machine learning workflows.

GitHub: github.com/DipeshGhimire33/Data-Science-and-Machine-learning

SOFT SKILLS

- **Team Collaboration:** Experience working in a 3-member development team and contributing to a shared codebase.
- **Problem Solving:** Hands-on experience troubleshooting database and schema-change issues during application development.
- **Adaptability & Learning:** Actively expanding from software development into Data Science and Machine Learning.
- **Communication:** Comfortable discussing technical challenges, asking questions, and working through problems collaboratively.
- **Curiosity:** Strong interest in understanding and applying emerging AI/ML technologies to practical problems.

CERTIFICATIONS

Kaggle Learn — Certificates of Completion

- **Intro to Programming** — Kaggle • September 1, 2026
- **Python** — Kaggle • September 2, 2026
- **Pandas** — Kaggle • September 3, 2026
- **Data Visualization** — Kaggle • September 3, 2026
- **Data Cleaning** — Kaggle • September 4, 2026
- **Feature Engineering** — Kaggle • September 6, 2026

Internship

- **Certificate of Experience — Machine Learning Data Analyst, Agribusiness** — YuvaIntern • September 7, 2026

Forage Job Simulations

- **GenAI Powered Data Analytics Job Simulation** — Tata (via Forage) • September 19, 2026
- **Data Analytics Job Simulation** — Deloitte (via Forage) • September 19, 2026
- **Data Labeling Job Simulation** — Forage • September 20, 2026
- **Data Science Job Simulation** — British Airways (via Forage) • September 20, 2026
- **Introduction to Data for Decision Makers Job Simulation** — BCG X (via Forage) • September 21, 2026